Supplementary material

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November 1, 2026

1 Appendix A. Reproduction

Every number in this draft is produced by code in scripts/ from artefacts hashed in reports/snapshot.json (230 runs, code, configs, splits, reports). The probes of §4.5 are reports/SATURATION_PROBE.md (pixel statistics), reports/hdr_render_probe.json (soft-shoulder re-render, script scripts/hdr_render_probe.py) and reports/M1_KILL_TEST.md (mask falsification, script scripts/m1_eval.py, per-image predictions in reports/m1_preds_s17,29.npz). The second renderer of §8.4 is reports/PHYSICAL_RENDERER.md, produced by scripts/physical_eval.py from src/maritime_robust/physical.py and gated by scripts/physical_gate.py; its pre-registration is reports/PREREG_PHYSICAL_RENDERER.md. A full re-evaluation of 12 runs reproduced the published R1 statistic bit-exactly (max absolute difference 0). The pre-registration for the severity 2/4 extension is in reports/PREREG_SEVERITY_2_4.md; the analysis and its comparison against the registered predictions is in reports/SEVERITY_CURVE.md. The error-overlap decomposition of §4.6 is reports/COMPOSITION.md §2, from scripts/overlap_main_grid.py; the fifteen-pair composition study and its replication gate are the rest of reports/COMPOSITION.md, from scripts/composition_calibrate,pairs,interaction,pr with the gate pre-registered in reports/PREREG_COMPOSITION_REPLICATION.md. All composition measurements are frozen-model inference; the only training added is six ResNet-50 clean baselines. The coverage study of §5 is pre-registered in reports/PREREG_COVERAGE.md (P1–P13, with the outcome of every prediction recorded there), calibrated by scripts/lattice_calibrate.py, run by scripts/coverage_lattice,singles,singles_hard,singles_mixbase_run.sh, and aggregated into reports/COVERAGE.md by scripts/coverage_aggregate.py; the figure is scripts/coverage_figure.py. The CIFAR-10-C replication of §4.7 is preregistered in reports/PREREG_CIFAR.md (C1–C5, B1–B3, A1–A3), trained by scripts/cifar_train.py (the AugMix baseline under A1–A3 is scripts/cifar_augmix.sh), evaluated over the 601-condition grid by scripts/cifar_eval.py, validated for port fidelity against the published archive by scripts/cifar_validate.py, and aggregated into reports/CIFAR_GRID.md by scripts/cifar_aggregate.py.

2 Appendix B. Margin-space superadditivity

The original analysis measured superadditivity in margin space (difference between compound accuracy and the additive expectation of margin losses): haze+glare extra margin loss -0.152 … -0.032, haze+cloud -0.144 … $+0.009$, across four backbone families. It agrees in sign with the accuracy-space result reported in §4.3 and is not used for inference here.

References